

```

sbit PWM=P2^4;
// PWM output
sbit RL1=P2^5;
// relay 1 pin
sbit RL2=P2^6;
// relay 2 pin

unsigned int x=10;          //
ontime                      //
unsigned int y=10;          //
offtime                     //
unsigned int m=0;           // mode
// mode
unsigned int d=0;           // direction
// direction
unsigned int t=100;         // time
// time
unsigned int r=0;           // run flag
// run flag

void start(void);
// function initialization
void mode(void);
void direction(void);
void incspeed(void);
void decspeed(void);
void incmode(void);
void decmode(void);
void time(unsigned int);
void delay(unsigned int);
void keydiy(void);
void busy(void);

void writemc(unsigned char a)
// send command to LCD
{
    busy();
    // check busy flag
    rs = 0;
    // select command register
    rw = 0;
    // write enable
    P0 = a;
    // send byte to LCD
    en = 1;
    // apply strobe pulse
    en = 0;
}

void writedata(unsigned char b)
// send data to LCD
{
    busy();
    // check busy flag
    rs = 1;
    // select data register
    rw = 0;
    // write enable
    P0 = b;
    // write enable
    en = 1;
    // send byte to LCD
    en = 0;
    // apply strobe pulse
}

void busy()
// check busy flag of
LCD
{
    en = 0;
    // disable display
    P0 = 0xFF;
    // P0 as input
    rs = 0;
    // select command register
    rw = 1;
    // read enable
    while(b==1)
        // if busy bit is 1
    {
        en=0;
        // remain within loop
        en=1;
    }
    en=0;
}

void writestr(unsigned char *s)
// send string message to LCD
{
    unsigned char i, l;
    l = strlen(s);
    // get length of string
    for(i=0; i<l; i++)
    {
        writedata(*s);
        // till the length of string
        s++;
        // send characters
        one by one
    }
}

void start()
// start rotating motor
{
    if(m==0)
        // for m=0 start continuous
    mode
    {
        RL1=0;
        // switch on RL1
        r=1;
        // set run flag
        Pl=0xFF;
        // send all 1's to P1
        while(Pl==0xFF)
            // till no key is pressed
        {
            led1=1;
            // indication on run LED
            PWM=1;
            // send high logic to PWM pin
            delay(x);
            // on time delay
            PWM=0;
            // now send low logic to PWM
            delay(y);
            // off time delay
        }
        // for m=1 start reversible mode
        {
            r=1;
            // set run flag
            Pl=0xFF;
            // send all 1's to P1
            while(Pl==0xFF)
                // till no key is pressed
            {
                led1=1; // run LED=1
                led3=1;
                led4=0;
                PWM=1; // send high on PWM pin
                RL2=1; // select one direction
                RL1=0; // switch on RL1
                time(t); // wait for desired time
                RL1=1; // switch off RL1
                led1=0; // run LED=0;
                time(20); // wait for 1 sec
                led1=1; // again run LED=1
                led3=0;
                led4=1;
                RL2=0; // select other direction
                RL1=0; // switch on RL1
                time(t); // wait for desire time
                RL1=1; // switch off RL1
                led1=0; // run LED=0
            }
        }
    }
}

time(20); // wait for 1 sec
}

PWM=0;
// for m=2 start jogging mode
{
    r=1;
    // reset run flag
    Pl=0xFF; // send all 1's to P1
    while(Pl==0xFF) // till no key is pressed
    {
        led1=1;
        PWM=1; // send high on PWM pin
        RL1=0; // switch on RL1
        time(t); // wait for 1 sec
        RL1=1; // switch off RL1
        PWM=0; // send low on PWM pin
        led1=0;
        time(20);
    }
}

void direction()
// alter the direction
{
    keydiy();
    // key debounce delay
    // increment count
    if((d&2)==0)
        // check for even or odd
    {
        led3=1;
        led4=0;
        RL2=1;
        // switch ON / OFF RL2
    }
    else
    {
        led3=0;
        led4=1;
        RL2=0;
    }
}

void mode()
// change mode of rotation
{
    keydiy();
    // key debounce delay
    writemc(0x80);
    // display message on first line
    first column
    m++;
    // increment count
    if(m==3) m=0;
    // if it is 3 reset it
    if(m==0)
    {
        writestr("mode:continuous");
        // otherwise display mode
        time(15);
    }
    else if(m==1)
    {
        writestr("mode:reversible");
        time(15);
    }
    else if(m==2)
    {
        writestr("mode:jogging");
        time(15);
    }
}

void decspeed()
// increase speed
{
}

```